Mixture-of-Experts Architecture

DeepSeek-V2

Efficient MoE Language Model with Multi-Head Latent Attention

236B	21B	128K
TOTAL PARAMS	ACTIVATED	CONTEXT

The LLM Scaling Dilemma

Larger models yield better intelligence but incur prohibitive training costs and slow inference due to KV cache bottlenecks.

Training Cost

Dense models require proportional compute for every parameter activated during training, making scale economically unfeasible.

KV Cache Bottleneck

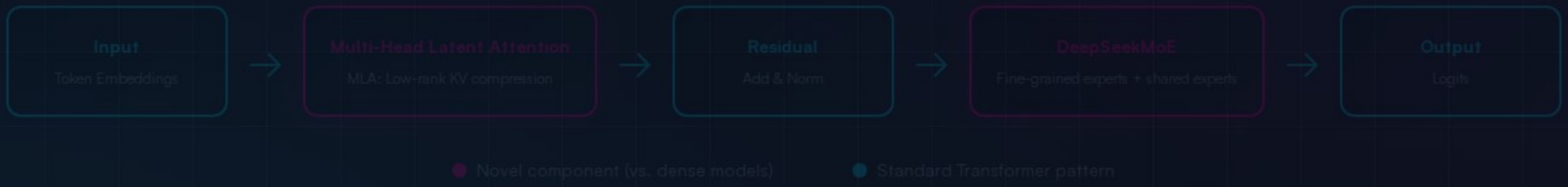
Standard attention stores full key-value pairs per head per token, creating memory walls during autoregressive generation.

Inference Speed

Larger dense models are slow to serve due to memory bandwidth constraints on activated parameters.

DeepSeek-V2 Architecture

Two key innovations within each Transformer block



Multi-Head Latent Attention

MHA

Full KV cache per head

N heads × d_head × 2

DeepSeekMoE: Fine-Grained Experts

Shared expert isolation + fine-grained segmentation for economical training

Shared Experts

- → Always activated for every token
- → Capture common knowledge across domains
- → Reduce redundancy in routing

Routed Experts

- → Top-K routing per token
- → Fine-grained expert segmentation (multiple small experts)
- → Increased specialization potential

Efficiency Gains vs. DeepSeek 67B

Relative improvements from architectural innovations

DeepSeek 67B

Dense · 67B activated

Training Cost	**Baseline**
KV Cache Size	**Baseline**
Generation Throughput	**Baseline**

DeepSeek-V2

MoE · 21B activated

Training Cost	**42.5% ↓**
KV Cache Size	**93.3% ↓**
Generation Throughput	**5.76× ↑**

Pareto-Optimal Efficiency

MMLU Performance: Pareto-Optimal Efficiency

Top-tier open-source MMLU score with only 21B activated parameters